Project: Worked examples
Title: Column base plate (EN 1993-1-8)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

An unstiffened rectangular base plate under an I/H, RHS or CHS column on grouted concrete: bearing resistance, the force regime and holding-down bolt tension for a major-axis moment, the tension T-stub of the plate and the compression T-stub, shear by friction and anchor bolts, and the plate thickness that satisfies them together. The bolt tension and shear per anchor go to Hilti PROFIS for the anchorage, which is done outside this sheet.
Bearing (cl. 6.2.5): f_cd = α_cc·f_ck/γ_c; f_jd = β_j·α·f_cd, β_j = 2/3; c = t_p·√[f_y/(3·f_jd·γ_M0)] Axial (cl. 6.2.8.2): N_j,Rd = f_jd·A_eff, A_eff the bearing strips of width c round the column footprint (full rectangle when they overlap); u_N = N_c,Ed/N_j,Rd Moment (cl. 6.2.8.3, Table 6.7): z_C from the section ((h − t_f)/2, (h − t)/2, half-annulus centroid) or entered, x_b the bolt row from the axis, z = z_C + x_b; C_Ed = (M_Ed + N_Ed·x_b)/z and T_Ed = C_Ed − N_Ed when both are ≥ 0; with N_Ed > 0 and T_Ed < 0 both sides are in compression, C = N_Ed/2 ± M_Ed/(2z_C); with N_Ed < 0 and C_Ed < 0 both rows are in tension, T = −N_Ed/2 ± M_Ed/(2x_b); F_t,Ed = T_Ed/n_t at the more loaded row Plate T-stub (cl. 6.2.6.11, 6.2.4, Table 6.2): M_pl,Rd = 0.25·l_eff·t_p²·f_y/γ_M0; without prying min(2·M_pl,Rd/m, n_t·F_t,Rd); with prying the modes 4·M_pl,Rd/m, (2·M_pl,Rd + n·ΣF_t,Rd)/(m + n) with n = min(e_min, 1.25m), and n_t·F_t,Rd; l_eff is the total for the n_t bolts Compression T-stub (cl. 6.2.6.9, 6.2.6.10): C_Rd = f_jd·A_flange, checked when M_Ed > 0 and the resultant is in compression Shear (cl. 6.2.2): F_2,vb,Rd = α_bc·f_ub·A_s/γ_M2, α_bc = 0.44 − 0.0003·f_yb; F_vb,Rd = min(F_1,vb,Rd, F_2,vb,Rd); V_Rd = C_f,d·N_c,Ed + n_b·F_vb,Rd; V_b,Ed = max(V_Ed − C_f,d·N_c,Ed, 0)/n_b Required thickness: the smallest t_p in 10–100 mm with u_N, u_C and u_T all ≤ 1 (each falls as t_p rises)

Ref: EN 1993-1-8:2005 cl. 6.2.2 — Design resistance of column bases in shear
Ref: EN 1993-1-8:2005 cl. 6.2.4, Table 6.2 — Equivalent T-stub in tension
Ref: EN 1993-1-8:2005 cl. 6.2.5 — Equivalent T-stub in compression
Ref: EN 1993-1-8:2005 cl. 6.2.6.9 — Concrete in compression including grout
Ref: EN 1993-1-8:2005 cl. 6.2.6.10 — Base plate in bending under compression
Ref: EN 1993-1-8:2005 cl. 6.2.6.11 — Base plate in bending under tension
Ref: EN 1993-1-8:2005 cl. 6.2.6.12 — Anchor bolts in tension
Ref: EN 1993-1-8:2005 cl. 6.2.8.2 — Column bases subject to axial forces
Ref: EN 1993-1-8:2005 cl. 6.2.8.3, Table 6.7 — Column bases subject to axial force and bending moment
Ref: EN 1992-1-1:2004 cl. 3.1.6, 6.7 — Design strength of concrete; partial area loading

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
column	Column section type	I/H		drawing: PF-GA-001 rev C	any
h	Column depth, along the plate length L (I/H and RHS)	260.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	50 … 1200 mm
b	Column width, along the plate width B (I/H and RHS)	256.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	50 … 1200 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
t_f	Flange thickness (I/H)	17.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	3 … 100 mm
t_w	Web thickness (I/H)	10.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	3 … 100 mm
B	Plate width	400	mm	drawing: PF-BP-001 rev B	100 … 1500 mm
L	Plate length	450	mm	drawing: PF-BP-001 rev B	100 … 1500 mm
t_p	Plate thickness	25	mm	drawing: PF-BP-001 rev B	10 … 100 mm
f_y	Plate yield strength	265	MPa	spec: BS EN 10025-2 S275, 16 mm < t ≤ 40 mm	235 … 460 MPa
f_ck	Concrete characteristic cylinder strength	30	MPa	spec: BS EN 206 C30/37 pad foundation, drawing PF-FN-001	20 … 50 MPa
f_ck_g	Grout characteristic strength (optional: checks the β_j = 2/3 condition)	60	MPa	spec: Non-shrink grout datasheet, 28-day strength (placeholder GR-DS-01)	5 … 100 MPa
t_g	Grout thickness (optional: checks the β_j = 2/3 condition)	30	mm	drawing: PF-BP-001 rev B	1 … 200 mm
alpha_cc	α_cc, long-term effects on the concrete strength	0.85		spec: UK NA to BS EN 1992-1-1, NA to cl. 3.1.6(1)P	0.8 … 1
gamma_c	γ_c, partial factor for concrete	1.5		spec: UK NA to BS EN 1992-1-1, Table NA.1	1.2 … 1.8
gamma_M0	γ_M0, partial factor for the plate	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.25
gamma_M2	γ_M2, partial factor for the anchor bolts	1.25		spec: UK NA to BS EN 1993-1-8, NA.2.3	1.1 … 1.5
C_fd	C_f,d, friction coefficient plate to grout	0.2		spec: EN 1993-1-8 cl. 6.2.2, sand-cement grout	0 … 0.45

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
alpha	α, concentration factor √(A_c1/A_c0) of the foundation (1 when the foundation geometry is not known)	1.5		assumption: Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_c1/A_c0 well above 2.25; checked on drawing PF-FN-001.	1 … 3
alpha_basis	Foundation geometry that α rests on (state it when α > 1)	Pad 1.2 × 1.2 m, 0.8 m deep (PF-FN-001): A_c1/A_c0 above 2.25, so α = 1.5 is taken		drawing: PF-FN-001 rev A	any
N_Ed	Axial force, COMPRESSION positive (the opposite sign to S3's N_Ed, which is tension positive; S2's R1_y, an upward reaction, is compression positive)	210	kN	reference: S5 load combinations sheet, ULS 6.10b, wind leading, column base A	-5000 … 10000 kN
M_Ed	Major-axis moment (uniaxial)	95	kN*m	reference: S5 load combinations sheet / S2 frame sheet, column base A	0 … 2000 kN*m
V_Ed	Shear force	60	kN	reference: S5 load combinations sheet / S2 frame sheet, column base A	0 … 2000 kN
n_t	Holding-down bolts in each bolt row (one row each side)	2		drawing: PF-BP-001 rev B	1 … 20
n_b	Holding-down bolts in total (both sides)	4		drawing: PF-BP-001 rev B	2 … 40
x_b	Bolt row distance from the column axis (outside the column face, inside the plate edge)	185	mm	drawing: PF-BP-001 rev B	50 … 1000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
m	Bolt to flange-weld toe distance, tension T-stub (required: it follows from x_b, the column and the weld)	46.9	mm	assumption: Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.	5 … 300 mm
l_eff	Effective length of the tension T-stub, total for the n_t bolts (required: from the cl. 6.2.6.5 patterns for the plate)	350	mm	assumption: Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).	10 … 2000 mm
prying	Prying forces develop (short bolts, L_b ≤ L_b*); off is conservative for the plate	no		assumption: Cast-in holding-down bolts: prying not assumed (min(2·M_pl/m, ΣF_t) is the conservative case).	any
F_t_Rd	Tension resistance per bolt (from S4)	203	kN	reference: S4 bolt sheet, M24 8.8 (203.04 kN), rounded down	1 … 2000 kN
f_ub	Anchor bolt ultimate strength	800	MPa	spec: EN ISO 898-1 class 8.8	400 … 1000 MPa
f_yb	Anchor bolt yield strength (α_bc is valid for 235–640 MPa)	640	MPa	spec: EN ISO 898-1 class 8.8	235 … 640 MPa
A_s	Anchor bolt tensile stress area (from S4)	352.5	mm^2	reference: S4 bolt sheet, M24 stress area (352.50 mm²)	50 … 1500 mm^2
F_1_vb_Rd	Bolt bearing resistance on the plate, F_1,vb,Rd (from S4 F_b,Rd)	364.4	kN	reference: S4 bolt sheet, M24 bearing on 25 mm S275 plate, e_1 = 60 mm, d_o = 27 mm (364.44 kN)	1 … 2000 kN
profis_ref	Hilti PROFIS report reference (the report is filed as Evidence)	PROFIS-BP1 (to be filed)		assumption: The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked.	any

3. Assumptions

A1. N_Ed is positive in compression here, the opposite of S3 (tension positive): a link from S2 must take the

reaction R1_y (up positive, compression into the base), never the member force N_e1. (Sign convention.)
A2. Static loading; a major-axis moment only (uniaxial); an unstiffened rectangular plate on grout of adequate strength and thickness for β_j = 2/3 (checked when the grout strength and thickness are entered). (Scope of S6.)
A3. The simplified T-stub model: one compression resultant at the centre of the compression T-stub (flange centre, or the half-annulus centroid of a CHS) and one tension resultant per bolt row; the force regime follows EN 1993-1-8 Table 6.7. The web is not given credit in the moment check. (EN 1993-1-8 cl. 6.2.8.3.)
A4. The effective length l_eff is the total for the n_t bolts of a row, read from the cl. 6.2.6.5 patterns; it is used for modes 1 and 2 alike (l_eff,1 ≤ l_eff,2, so this is conservative for mode 2). (One input.)
A5. Concrete, anchor tension, pull-out, breakout, edge and the tension–shear interaction of the anchors are designed in Hilti PROFIS from F_t,Ed and V_b,Ed per anchor; the report is filed as Evidence and referenced. With prying switched on the prying force Q is not in F_t,Ed and PROFIS must add it. (External anchorage design.)
A6. The friction resistance counts only the compression, C_f,d·max(N_Ed, 0); a column in net tension has none. Anchor bolts share shear equally, with their holes assumed no larger than the standard clearance (oversized or slotted holes are outside this sheet). (EN 1993-1-8 cl. 6.2.2.)
A7. The bearing outstand c is cut back where it would pass the plate edge; the web and the column interior keep the full c. The CHS compression side is half of the annular bearing area, with its centroid as lever arm (an approximation). (Conservative against an uncut c.)
A8. The compression T-stub is checked against the flange area alone as soon as M_Ed > 0, while at M_Ed = 0 the whole footprint (web included) resists, so the compression check jumps at M_Ed = 0^{+} (conservative). (Limitation.)
A9. Partial factors and the friction coefficient come from the records in use (γ_c, α_cc: EN 1992-1-1 NA; γ_M2: EN 1993-1-8 NA; γ_M0: EN 1993-1-1 NA; C_f,d: the grout used); the values shown default to the EN recommended and UK values. (§A5 small factor set.)
A10. alpha (α, concentration factor √(A_c1/A_c0) of the foundation (1 when the foundation geometry is not known)) is taken as 1.5. (Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_c1/A_c0 well above 2.25; checked on drawing PF-FN-001.)
A11. m (Bolt to flange-weld toe distance, tension T-stub (required: it follows from x_b, the column and the weld)) is taken as 46.9 mm. (Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.)
A12. l_eff (Effective length of the tension T-stub, total for the n_t bolts (required: from the cl. 6.2.6.5 patterns for the plate)) is taken as 350 mm. (Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).)
A13. prying (Prying forces develop (short bolts, L_b ≤ L_b*); off is conservative for the plate) is taken as no. (Cast-in holding-down bolts: prying not assumed (min(2·M_pl/m, ΣF_t) is the conservative case).)
A14. profis_ref (Hilti PROFIS report reference (the report is filed as Evidence)) is taken as PROFIS-BP1 (to be filed). (The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked.)

4. Working

`[31] f_cd = alpha_cc * f_ck / gamma_c` EN 1992-1-1 cl. 3.1.6

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= 0.85 × 30 MPa / 1.5
= 17 MPa
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Concrete design compressive strength

`[32] beta_j = 2/3, fixed` EN 1993-1-8 cl. 6.2.5

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= grout conditions of cl. 6.2.5 assumed met
= 0.66667
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Joint coefficient β_j

[33] f_jd = beta_j * alpha * f_cd EN 1993-1-8 cl. 6.2.5
= 0.66667 × 1.5 × 17 MPa
= 17 MPa
Design bearing strength of the joint
[34] c = t_p * sqrt(f_y / (3 * f_jd * gamma_M0)) EN 1993-1-8 cl. 6.2.5
= 25 mm × sqrt(265 MPa / (3 × 17 MPa × 1))
= 56.987 mm
Additional bearing width round the column footprint
[35] A_eff = 2(b + 2c)(t_f + 2c) + (h - 2t_f - 2c)(t_w + 2c) EN 1993-1-8 cl. 6.2.5
= 2 × 370.27 × 131.27 + 111.73 × 124.27 mm²
= 111100 mm²
Effective bearing area
[38] N_j_Rd = f_jd * A_eff EN 1993-1-8 cl. 6.2.8.2, 6.2.5
= 17 MPa × 111100 mm²
= 1888.7 kN
Axial bearing resistance of the joint
[39] N_c_Ed = max(N_Ed, 0)
= N_Ed = 210 kN
= 210 kN
Design compression in the column
[40] u_N = N_c_Ed / N_j_Rd
= 210 kN / 1888.7 kN
= 0.11119
Axial bearing utilisation
[41] t_req_N = smallest t_p in 10–100 mm with f_jd·A_eff(c(t_p)) ≥ N_c,Ed (bisection) EN 1993-1-8 cl. 6.2.5
= N_c,Ed = 210 kN; carried at the 10 mm minimum
= 10 mm
Plate thickness required for axial bearing alone
[42] z_C = (h - t_f)/2 EN 1993-1-8 cl. 6.2.8.3
= (260.3 - 17.3)/2 mm
= 121.5 mm
Compression resultant distance from the column axis
[43] z = z_C + x_b EN 1993-1-8 cl. 6.2.8.3
= 121.5 + 185
= 306.5 mm
Lever arm between the compression resultant and the bolt row
[44] C_Ed = (M_Ed + N_Ed·x_b)/z EN 1993-1-8 cl. 6.2.8.3, Table 6.7
= (95 kN·m + 210 kN × 185 mm)/306.5 mm
= 436.7 kN
Compression resultant under the compression flange
[45] T_Ed = C_Ed - N_Ed EN 1993-1-8 cl. 6.2.8.3, Table 6.7
= 436.7 - 210
= 226.7 kN
Tension resultant at the bolt row
[46] C_Ed_2 = none
= other side
= 0 kN
Compression resultant under the other flange
[47] T_Ed_2 = none
= other row
= 0 kN
Tension resultant at the other bolt row

[48] F_t_Ed = T_Ed / n_t — EN 1993-1-8 cl. 6.2.6.12
= 226.7 kN / 2
= 113.35 kN
Holding-down bolt tension per bolt at the more loaded row (→ PROFIS)
[49] A_flange = (b + 2c)(t_f + 2c), at most A_eff/2 — EN 1993-1-8 cl. 6.2.6.9
= min(370.27 × 131.27, A_eff/2) mm²
= 48608 mm²
Bearing area of the compression flange
[50] C_Rd = f_jd * A_flange — EN 1993-1-8 cl. 6.2.6.9, 6.2.6.10
= 17 MPa × 48608 mm²
= 826.33 kN
Bearing resistance of the compression flange
[51] u_C = C_Ed / C_Rd
= 436.7 kN / 826.33 kN
= 0.52849
Compression flange utilisation
[52] M_pl_Rd = 0.25 * l_eff * t_p^2 * f_y / gamma_M0 — EN 1993-1-8 Table 6.2
= 0.25 × 350 mm × (25 mm)^2 × 265 MPa / 1
= 14.492 kN·m
Plastic moment resistance of the plate over l_eff
[53] T_Rd_1 = 2 * M_pl_Rd / m — EN 1993-1-8 Table 6.2
= 2 × 14.492 kN·m / 46.9 mm
= 618 kN
Mode 1-2 without prying forces
[54] T_Rd_3 = n_t * F_t_Rd — EN 1993-1-8 Table 6.2
= 2 × 203 kN
= 406 kN
Mode 3: bolt failure
[55] T_Rd = min(T_Rd_1, T_Rd_3) — EN 1993-1-8 cl. 6.2.4
= min(618 kN, 406 kN)
= 406 kN
Tension T-stub resistance (no prying)
[56] u_T = T_Ed / T_Rd
= 226.7 kN / 406 kN
= 0.55839
Plate bending and bolts in tension, T-stub utilisation
[57] u_bt = F_t_Ed / F_t_Rd
= 113.35 kN / 203 kN
= 0.55839
Holding-down bolt tension utilisation
[58] t_req = smallest t_p in 10–100 mm with u_N, u_C, u_T ≤ 1 (bisection) — EN 1993-1-8 cl. 6.2.5, 6.2.6.9, 6.2.6.11
= N_c,Ed = 210 kN, C_Ed = 436.7 kN, T_Ed = 226.7 kN
= 15.142 mm
Plate thickness required
[59] alpha_bc = 0.44 - 0.0003·f_yb — EN 1993-1-8 cl. 6.2.2
= 0.44 - 0.0003 × 640
= 0.248
Anchor bolt shear factor α_bc
[60] F_2_vb_Rd = alpha_bc * f_ub * A_s / gamma_M2 — EN 1993-1-8 cl. 6.2.2
= 0.248 × 800 MPa × 352.5 mm² / 1.25
= 55.949 kN
Anchor bolt shear resistance F_2,vb,Rd

[61] F_vb_Rd = min(F_1_vb_Rd, F_2_vb_Rd) EN 1993-1-8 cl. 6.2.2
= min(364.4 kN, 55.949 kN)
= 55.949 kN
Anchor bolt shear resistance per bolt (bearing or shear)
[62] F_f_Rd = C_fd * N_c_Ed EN 1993-1-8 cl. 6.2.2
= 0.2 × 210 kN
= 42 kN
Friction resistance, compression only
[63] V_Rd = F_f_Rd + n_b * F_vb_Rd EN 1993-1-8 cl. 6.2.2
= 42 kN + 4 × 55.949 kN
= 265.8 kN
Shear resistance of the base
[64] u_V = V_Ed / V_Rd
= 60 kN / 265.8 kN
= 0.22574
Shear utilisation
[65] V_b_Ed = max(V_Ed - F_f,Rd, 0)/n_b EN 1993-1-8 cl. 6.2.2
= max(60 - 42, 0)/4
= 4.5 kN
Shear per anchor bolt after friction (→ PROFIS)
[66] u_max = max(u_N, u_C, u_T, u_bt, u_V)
= u_N = 0.11119, u_C = 0.52849, u_T = 0.55839, u_bt = 0.55839, u_V = 0.22574
= 0.55839
Governing utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
f_cd	17 MPa	Concrete design strength
beta_j	0.66667	Joint coefficient β_j
f_jd	17 MPa	Bearing strength f_jd
c	56.987 mm	Bearing outstand c
A_eff	111100 mm²	Effective bearing area
N_j_Rd	1888.7 kN	Axial bearing resistance
u_N	0.11119	Axial bearing utilisation
t_req_N	10 mm	Plate thickness required for axial bearing alone
t_req	15.142 mm	Plate thickness required (bearing, compression and tension T-stubs)
z_C	121.5 mm	Compression lever arm z_C
z	306.5 mm	Lever arm z
C_Ed	436.7 kN	Compression resultant
C_Ed_2	0 kN	Compression resultant, other side
T_Ed	226.7 kN	Tension resultant, more loaded row
T_Ed_2	0 kN	Tension resultant, other row
F_t_Ed	113.35 kN	Bolt tension per bolt (to PROFIS)
A_flange	48608 mm²	Compression-side bearing area

SYMBOL	VALUE	DESCRIPTION
C_Rd	826.33 kN	Compression-side resistance
u_C	0.52849	Compression-side utilisation
M_pl_Rd	14.492 kN·m	Plate plastic moment
T_Rd_1	618 kN	T-stub mode 1-2, no prying
T_Rd_3	406 kN	T-stub mode 3
T_Rd	406 kN	Tension T-stub resistance
u_T	0.55839	Tension T-stub utilisation
u_bt	0.55839	Bolt tension utilisation
alpha_bc	0.248	Anchor shear factor α_bc
F_2_vb_Rd	55.949 kN	Anchor shear resistance F_2,vb,Rd
F_vb_Rd	55.949 kN	Anchor shear resistance per bolt
F_f_Rd	42 kN	Friction resistance
V_Rd	265.8 kN	Shear resistance
u_V	0.22574	Shear utilisation
V_b_Ed	4.5 kN	Shear per anchor bolt (to PROFIS)
u_max	0.55839	Governing utilisation
regime	"tension at one bolt row, compression under the other flange (Table 6.7, z = z_C + x_b)"	Force regime (Table 6.7)
T_mode	"mode 3 (bolt failure)"	Governing T-stub mode

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Base plate tension T-stub (bending at the bolts)	u_T = 0.55839	u_T ≤ 1	0.558	EN 1993-1-8 cl. 6.2.6.11
PASS	Holding-down bolt tension	u_bt = 0.55839	u_bt ≤ 1	0.558	EN 1993-1-8 cl. 6.2.6.12
PASS	Compression flange bearing	u_C = 0.52849	u_C ≤ 1	0.528	EN 1993-1-8 cl. 6.2.6.9, 6.2.6.10
PASS	Shear: friction plus anchor bolts	u_V = 0.22574	u_V ≤ 1	0.226	EN 1993-1-8 cl. 6.2.2
PASS	Concrete bearing under the axial force	u_N = 0.11119	u_N ≤ 1	0.111	EN 1993-1-8 cl. 6.2.8.2, 6.2.5

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: alpha_cc, gamma_c (EN 1992-1-1 National Annex factors); gamma_M2 (EN 1993-1-8 National Annex factor); gamma_M0 (EN 1993-1-1 National Annex factor); C_fd (Friction coefficient plate to grout); f_y (Plate strength). Enter each as one so its source, edition and licence travel with it.

WARNING — anchorage: profis_ref: 'PROFIS-BP1 (to be filed)' is typed text, not an attached report; attach the Hilti PROFIS report as Evidence so the reference points at the filed document. The anchorage design is not checked on this sheet.

8. Verification

Template struct.base-plate@1.1.0 Content hash ceb71fd8f23f78af Engine calc-engine/1.0.0
Known-answer cases passed 9/9 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0